

Answer of Intermediate Microeconomics Practice 1

Mochiao Chen

January 2, 2026

Instructions:

- Answer all questions in English.
- Write your calculation process clearly for computational problems.
- For graphing questions, clearly label all axes, curves, and equilibrium points.

Part I: True or False (20 points)

Indicate whether the following statements are True (T) or False (F). Briefly explain your reasoning for any False statements.

1. ☐ Walras' Law states that if $n - 1$ markets in an exchange economy are in equilibrium, the n -th market must also be in equilibrium.
2. ☐ In a First-Degree (Perfect) Price Discrimination scenario, the monopolist extracts all consumer surplus, resulting in zero deadweight loss.
3. ☐ A pure public good is characterized by being rival in consumption and excludable.
4. ☐ In the Prisoner's Dilemma game, the Nash Equilibrium is always Pareto efficient.
5. ☐ Adverse selection usually occurs *before* the transaction takes place due to hidden information, whereas moral hazard occurs *after* the transaction due to hidden action.
6. ☐ Under Third-Degree Price Discrimination, a monopolist will charge a higher price in the market with more elastic demand (more negative price elasticity).
7. ☐ According to the Coase Theorem, if property rights are well-defined and transaction costs are zero, the initial allocation of property rights determines whether the efficient outcome can be achieved.
8. ☐ The "Core" of an exchange economy is the set of all Pareto-optimal allocations that are welfare-improving for both consumers relative to their initial endowments.
9. ☐ In a natural monopoly, setting the price equal to Marginal Cost ($P = MC$) allows the firm to break even (zero economic profit).
10. ☐ In a mixed-strategy Nash Equilibrium, a player chooses probabilities such that the opponent is indifferent between their pure strategies.

Part II: Multiple Choice (20 points)

Select the best answer for each question.

1. A monopolist faces a demand curve with constant elasticity $\varepsilon = -3$. If the marginal cost of production is constant at \$20, what is the profit-maximizing price?
 - A. \$20
 - B. \$30
 - C. \$40
 - D. \$60
2. Which of the following best describes the "Tragedy of the Commons"?
 - A. A failure of public goods provision due to free-riding.
 - B. Over-consumption of a non-excludable but rival resource.
 - C. Under-production of a good with positive externalities.
 - D. A coordination failure in a Stag Hunt game.
3. In an Edgeworth Box, the Contract Curve consists of:
 - A. All allocations where one consumer is better off than the other.
 - B. All allocations where the marginal rate of substitution (MRS) is equal for both consumers.
 - C. All allocations inside the "lens" formed by the initial indifference curves.
 - D. Only the competitive equilibrium points.
4. Consider the "Market for Lemons" (Adverse Selection). If buyers cannot distinguish between high-quality and low-quality cars, and only know the average value:
 - A. Only high-quality cars will be traded.
 - B. Both types will be traded at the average price.
 - C. High-quality car sellers may exit the market, leading to a market collapse.
 - D. Signaling has no effect on this market.
5. The efficiency condition for the provision of a public good with two consumers (A and B) is given by:
 - A. $MRS_A = MRS_B = MRT$
 - B. $MRS_A + MRS_B = MC$ (or MRT)
 - C. $MU_A = MU_B$
 - D. $MRS_A/P_A = MRS_B/P_B$
6. In a Sequential Game where Firm A moves first (Stackelberg leader) and Firm B moves second (follower):
 - A. Firm A usually achieves a higher profit than in a simultaneous Cournot game.

- B. Firm B has the "first-mover advantage."
 - C. The outcome is identical to the Cournot Nash Equilibrium.
 - D. Both firms produce the monopoly quantity.
7. If a monopolist implements a Two-Part Tariff ($T = F + p \cdot q$) to maximize total profit and extract all surplus from identical consumers, it should set the per-unit price p equal to:
- A. The monopoly price.
 - B. Zero.
 - C. Marginal Cost.
 - D. Average Total Cost.
8. High-ability workers getting a college degree to distinguish themselves from low-ability workers, even if the degree adds no productivity, is an example of:
- A. Moral Hazard
 - B. Screening
 - C. Signaling
 - D. Bundling
9. In the presence of a negative production externality (e.g., pollution), the private market equilibrium quantity is usually _____ than the socially optimal quantity.
- A. Lower
 - B. Higher
 - C. The same
 - D. Unrelated
10. A situation where a consumer's utility depends on the consumption of others, or a production function depends on the activities of others, without market compensation, is defined as:
- A. An Externality
 - B. A Public Good
 - C. Imperfect Competition
 - D. Asymmetric Information

Part III: Term Definitions (15 points)

Briefly explain the following terms in the context of microeconomics (2-3 sentences each).

1. **Pareto Efficiency**
2. **Nash Equilibrium**
3. **Moral Hazard**
4. **Third-Degree Price Discrimination**
5. **Free-Rider Problem**

Part IV: Calculations (30 points)

Show all steps of your work.

Question 1: Monopoly and Price Discrimination (10 points)

A monopolist sells a product in two separate markets, Market 1 and Market 2. The demand curves are:

$$P_1 = 100 - Q_1$$

$$P_2 = 80 - 2Q_2$$

The monopolist's total cost function is $TC(Q) = 20Q$, where $Q = Q_1 + Q_2$.

- (a) If the monopolist can practice third-degree price discrimination, calculate the profit-maximizing quantity (Q_1, Q_2) and price (P_1, P_2) for each market.
- (b) Calculate the total profit under price discrimination.
- (c) Which market has a more elastic demand at the equilibrium point? Verify using the markup rule or elasticity formula.

Question 2: Game Theory and Mixed Strategies (10 points)

Consider the following payoff matrix for a game between Player A and Player B:

	B: Left	B: Right
A: Up	2, 3	0, 1
A: Down	1, 0	3, 2

- (a) Does this game have any pure-strategy Nash Equilibria? If yes, identify them. If no, explain why.
- (b) Find the mixed-strategy Nash Equilibrium. Let p be the probability Player A plays Up, and q be the probability Player B plays Left. Calculate the equilibrium values for p and q .

Question 3: Externalities (10 points)

A steel factory (Firm S) is located upstream from a fishery (Firm F). The steel firm's profit function is $\pi_S = 12s - s^2 - (x - 4)^2$, where s is steel output and x is pollution. The fishery's profit function is $\pi_F = 10f - f^2 - xf$, where f is fish output. The pollution x negatively affects the fishery.

- (a) If the firms operate independently (steel firm ignores the external cost), what level of pollution x will the steel firm choose?
- (b) If the two firms merge and maximize joint profit, what is the socially optimal level of pollution x^* ?
- (c) Comparing (a) and (b), does the independent market produce too much or too little pollution?

Part V: Graphing and Analysis (15 points)

Topic: General Equilibrium and the Edgeworth Box

Consider an exchange economy with two consumers, A and B, and two goods, 1 and 2. Consumer A has an initial endowment $\omega_A = (8, 2)$ and Consumer B has an initial endowment $\omega_B = (2, 8)$. Their preferences are standard (convex indifference curves).

- (a) Draw an **Edgeworth Box** diagram. Clearly label the following:
- The dimensions of the box (total quantity of Good 1 and Good 2).
 - The endowment point W .
 - An indifference curve for A and an indifference curve for B passing through the endowment point.
- (b) On your diagram, shade the area representing the **Lens of Trade** (the set of allocations that make both consumers better off).
- (c) Illustrate and label the **Contract Curve**. Explain what condition holds true at every point on the Contract Curve.
- (d) Identify the **Core** on the Contract Curve.
- (e) Draw a budget line and a potential competitive equilibrium point. Explain how the price ratio (P_1/P_2) relates to the indifference curves at this equilibrium.